Status	Finished
Started	Tuesday, 20 October 2026, 7:05 PM
Completed	Tuesday, 20 October 2026, 7:10 PM
Duration	4 mins 49 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 1000
- b. 500
- c. 10 ✓
- d. 100

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(n log n)
- b. O(log n) ✓
- c. O(1)
- d. O👎

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O(log n)
- c. O👎 ✓
- d. O(n log n)

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search can find multiple instances of the target element
- b. Binary search works on unsorted lists
- c. Binary search does not require dividing the list
- d. Binary search has a lower time complexity on large, sorted lists ✓

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search continues in the left sublist
- b. The search ends successfully ✓
- c. The search continues in the right sublist
- d. The list is sorted

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By using a hash function
- b. By dividing the list length by two ✓
- c. By starting from the first element
- d. By comparing each element sequentially

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the search algorithm finishes
- c. When the list contains duplicate elements
- d. When the element is found in the list ✓

Question **8**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. Item is the last element in the array or item is not there at all
- c. When the item is the last element in the array
- d. When the item is not the array at all

Question **9**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. In the right sublist ✓
- c. In the left sublist
- d. At the beginning of the list

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 94 and 99
- b. 90 and 99 ✓
- c. 89 and 94
- d. 90 and 100

Question **11**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n ✓
- c. 1
- d. n/2

Question **12**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Hash search
- d. Both (1) & (3)

Question **13**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It works on unsorted lists
- b. It always starts from the beginning of the list
- c. It can be applied only if the list is sorted ✓
- d. It compares elements sequentially

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The last element is returned
- b. An error is raised
- c. The first element is returned
- d. The search is considered unsuccessful ✓

Question **15**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n/2
- b. n-1
- c. log n
- d. n ✓